

Ciem Cornelissen

PhD Researcher in Computer Science Engineering

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EDUCATION

Ghent University - imec <i>PhD in Computer Science Engineering</i>	Ghent, Belgium Jun. 2024 – Present
KU Leuven <i>Advanced Master of Artificial Intelligence – Magna Cum Laude</i>	Leuven, Belgium Sep. 2022 – Jun. 2023
Ghent University <i>Master of Science in Physics – Cum Laude</i>	Ghent, Belgium Sep. 2020 – Jun. 2022

RESEARCH EXPERIENCE

PhD Researcher <i>IDLab, Ghent University - imec</i>	Jun. 2024 – Present Ghent, Belgium
- Developed Le MuMo JEPA, extending joint-embedding predictive architectures to multi-modal perception through learnable fusion tokens and achieving the strongest performance-efficiency trade-off among from-scratch multi-modal baselines (CVPR 2026 URVIS) - Developed LaFreeWM, a Label-Free action-conditioned JEPA world model that infers discrete latent actions during action-label-free pretraining and uses 1% or 10% labelled trajectories for continuous-action planning - Developed LISA, a domain-adversarial deep learning framework for illumination-invariant multi-modal analysis, improving prediction generalization by over 20% across varying environmental conditions - Created the OHSLIC algorithm for efficient on-device sensor segmentation, achieving 10x inference speedup (12.2ms vs 130.2ms) on an NVIDIA Jetson Nano for real-time embedded perception	

PUBLICATIONS

LaFree World Model: Label-Free Pretraining for Action-Conditioned JEPA World Models	2026
<i>Manuscript under review at ICLR 2027</i> (submitted Sep. 2026) - Infers discrete latent actions from unlabelled observations, then grounds the world model with limited labelled data for continuous-action planning - Authors: C. Cornelissen, S. Leroux, P. Simoens preprint	
Le MuMo JEPA: Multi-Modal Self-Supervised Representation Learning	2026
<i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) – URVIS Workshop</i> (Jun. 2026) - Self-supervised framework learning unified representations from RGB and companion modalities via learnable fusion tokens - Authors: C. Cornelissen, S. Leroux, P. Simoens Paper Code	
In-Field Mapping of Grape Yield and Quality with Illumination-Invariant Deep Learning	2025
<i>IEEE Internet of Things Journal</i> (Oct. 2025) - Developed LISA framework for illumination-invariant hyperspectral imaging in precision viticulture - Authors: C. Cornelissen, S. De Coninck, A. Willekens, S. Leroux, P. Simoens Paper Code	
Adaptive Clustering for Efficient Phenotype Segmentation of UAV Hyperspectral Data	2025
<i>IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops</i> (Mar. 2025) - Introduced OHSLIC algorithm for computationally efficient, real-time phenotype segmentation - Authors: C. Cornelissen, S. Leroux, P. Simoens Paper Code	

TECHNICAL SKILLS

Research Interests: Self-Supervised Learning, Predictive Representations, World Models, JEPA, Multi-Modal Learning, Computer Vision
Perception: RGB/LiDAR/Thermal Fusion, Depth Estimation, Semantic Segmentation, 3D Object Detection, On-Device Inference
Programming & Tools: Python, PyTorch, C/C++, TensorFlow, Git, Docker, Linux, L^AT_EX
Languages: Dutch (Native), English (Fluent, C2 level)